IDLCV Project II: Video Classification
Group 26

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1. Introduction

The goal of this project is to develop models capable of classifying video clips based on their action content. Specifically, we work with a subset of the UCF-101 dataset containing 10 workout-related action classes, where each video is represented by 10 uniformly sampled frames. To achieve this, we explore multiple architectural approaches for video understanding, starting with simple per-frame aggregation methods and progressing to more sophisticated temporal modeling techniques including late fusion, early fusion, and 3D convolutional networks. We also investigate two-stream architectures that leverage both RGB frames and optical flow to capture appearance and motion information separately. A critical component of this work addresses the problem of data leakage between training, validation, and test splits, where the same subjects appeared across splits in the original dataset. We retrain selected models on corrected splits to evaluate their true generalization performance and assess the impact of proper split construction on model evaluation.

2. Architecture

We implemented several architectures for video classification, starting from per-frame aggregation models and extending to temporal fusion and 3D CNNs.

2.1. Aggregation of per-frame models

In this part of the project, every frame was processed separately by a 2D convolutional neural network (CNN), and the predictions from all frames were then averaged to get one final prediction for the whole video. The model was trained using the UCF-101 dataset with the Adam optimizer and CrossEntropyLoss for 10 epochs.

The 2D aggregation model reached a best validation accuracy of 87.3% and a test accuracy of 86.7%. The training was stable, and the model was able to recognize most actions correctly. This shows that a 2D CNN can already perform well for actions that are easy to identify from single frames, such as JumpingJack or HandstandPushup.

A 3D version of the aggregation model was also tested, using 3D convolutions to learn both spatial and temporal features at the same time. The results using the 3D convolutional network were a test loss of 1,45 and a test accuracy of 63,5. We can appreciate that there is a drop in performance.

2.2. Late fusion Models

We developed two versions of Late Fusion Models - a 2D and a 3D one. Both models consist of a backbone stage (composed of respectively 2D and 3D stacked convolutional blocks with batch normalization, ReLU activations, and pooling layers) followed by a fusion phase aimed at aggregating temporal information for video classification. The 2D version, based on frame-by-frame convolutional processing with temporal feature fusion, achieved the best results with a test accuracy of 90% and a loss of 0.3381, demonstrating strong generalization thanks to regularization techniques such as early stopping and dropout. The 3D version, on the other hand, uses spatiotemporal convolutions to jointly capture motion and appearance cues, obtaining a test accuracy of 81.7% and a loss of 0.6051. While 3D convolutions have greater temporal modeling capacity in principle, our 2D late-fusion model achieved higher test accuracy in this study.

2.3. Early fusion

Our initial experiments with early fusion on the original dataset yielded surprisingly high performance, achieving 96.67% test accuracy with the 2D approach (conv1x1 mixing). However, we observed that training accuracy reached 100%, and notably, test accuracy exceeded validation accuracy (94.17%), which raised concerns about the experimental setup. Interestingly, when we implemented early fusion with 3D convolutions (conv3d with kernel size 3×1×1), performance decreased to 87.50% test accuracy despite the model having more capacity for temporal reasoning.

2.4. Summary of the models

We summarized the results of all models in the next table:

Model	Test Loss	Test Acc (%)
2D Aggregation	1.02	86.7
3D Aggregation	1.45	63.5
2D Late Fusion	0.34	90.0
3D Late Fusion	0.61	81.7
2D Early Fusion	0.64	96.7
3D Early Fusion	0.83	87.5

Table 1. Test loss and accuracy for all six models.

Overall, the results are high, getting in most cases more than 80% accuracy, except for the accuracy of the 3D aggregation model. We can see that in the three different approaches there is a drop in performance when we use 3D convolutions instead of the more basic 2D ones. One reason behind this drop could be that 3D convolutions need more data to get better results.

Upon investigation, we identified data leakage in the original Kaggle-derived dataset splits. The same individuals appeared across training, validation, and test sets, meaning the model could exploit person-specific appearance features (facial characteristics, body shape, clothing) rather than learning generalizable action patterns. This explains why the simpler appearance-based 2D method outperformed the more complex 3D temporal modeling approach - with leakage, the task reduces to person recognition rather than action recognition, and additional complexity only introduces overfitting.

The counterintuitive result that simpler models outperform more sophisticated temporal models serves as a strong indicator of data leakage. When person identity is predictive of class labels, methods that focus on appearance (2D) excel, while methods attempting to learn motion patterns (3D) struggle unnecessarily with a harder objective than required.

To address this issue, we retrained our the model with best results, 2D Early Fusion, using the corrected `ucf101noleakage` dataset, where individuals are strictly separated across splits.

3. Information leakage between train/val/test split

To quantify the impact of the data leakage identified, we retrained our best-performing 2D Early Fusion model on the corrected `ucf101_noleakage` dataset. This dataset ensures that subjects from the training set do not appear in the validation or test sets.

As hypothesized, this correction resulted in a dramatic drop in performance, confirming that our original model was heavily overfitting to subject-specific features rather than generalizing to the actions themselves. The model's test accuracy fell from **96.67%** on the original dataset to **56.67%** on the 'no-leakage' dataset. The best validation accuracy achieved was **47.5%** with a final test loss of **1.6686**. This new, lower accuracy serves as a far more realistic and trustworthy baseline for our model's true generalization capability.

Analysis of the new confusion matrix (Figure 1) reveals the model's specific challenges. While it still performs well on distinct actions like `Lunges` (10/12) and `JumpRope` (8/12), it now struggles with semantically similar classes. The most significant confusion occurs between `JumpingJack` and `BodyWeightSquats`, with 3 `JumpingJack` videos being misclassified as `BodyWeightSquats`. Similarly, 4 `HandstandWalking` videos were misclassified as `Lunges`, and 4 `PushUps` were mistaken for `TrampolineJumping`. There is also predictable confusion between similar vertical-pushing motions, such as `WallPushups` being mistaken for `HandstandPushups` (3 instances) and `PullUps` for `WallPushups` (3 instances).

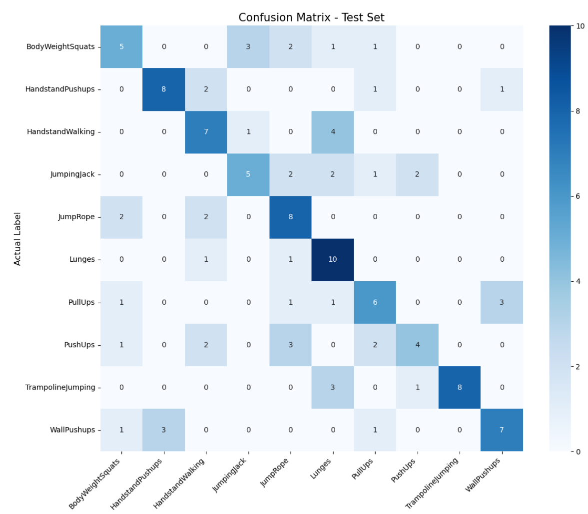

Figure 1. Confusion matrix for the 2D Early Fusion model on the `ucf101_noleakage` test set.

These results underscore the critical importance of proper dataset splitting. The 56.67% accuracy baseline clearly illustrates the difficulty of genuine action recognition based on RGB frames alone, setting the stage for more advanced models, like the two-stream network, which are designed to tackle this more difficult and realistic task.

4. Dual Stream Network

For the last part of the project, we implemented the Dual Stream Network model. This model leverages the temporal aspect of the optical flows paired with the analysis of each frame individually.

The model implemented has the architecture of the figure 2 on the project description, that is the architecture proposed by Simonyan and Andrew Zisserman in Two-stream architecture for video classification [3]. In order to get the same architecture, the resize had to be set to 256x256 pixels. We also added other transformations as horizontal flip, color jitter and normalization after getting the tensor of the frames. The training employs the UCF101noleakage dataset, avoiding the problems described before but also suffering the drop of performance. Also, it was optimized using *Adam (lr = 1e-4)* and *CrossEntropyLoss* with early stopping to avoid overfitting.

The final results of the training are a test accuracy of **53,33%** and a test loss of **1.897**. Next figure shows the training loss. It is worth mentioning that due to the early stopping it only needed 4 epochs to get the best validation loss. We tried with higher patience but the results were not different. Also the results are not as high as expected, getting even worse results than using Early Fusion.

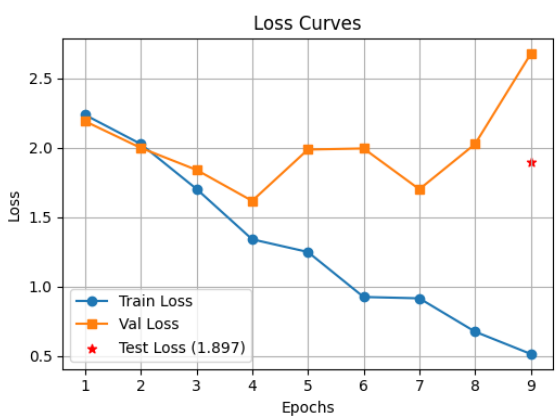

Figure 2. Train and validation loss

We also analyzed the confusion matrix to examine misclassifications made by the model. Notably, the model struggled to correctly classify any of the `Lunges` videos, often confusing them with `JumpRope`. On the other hand, it performed well on classes such as `HandstandPushups`, `JumpingJack`, `Lunges`, and `Push Ups`, achieving high accuracy in these categories. For the remaining classes, the model achieved approximately half of the predictions correct. Some confusions, like those between `HandstandPushups` and `HandstandWalking`, may be explained by similarities similar flows, for example the vertical body position, while other misclassifications do not appear to have an obvious justification.

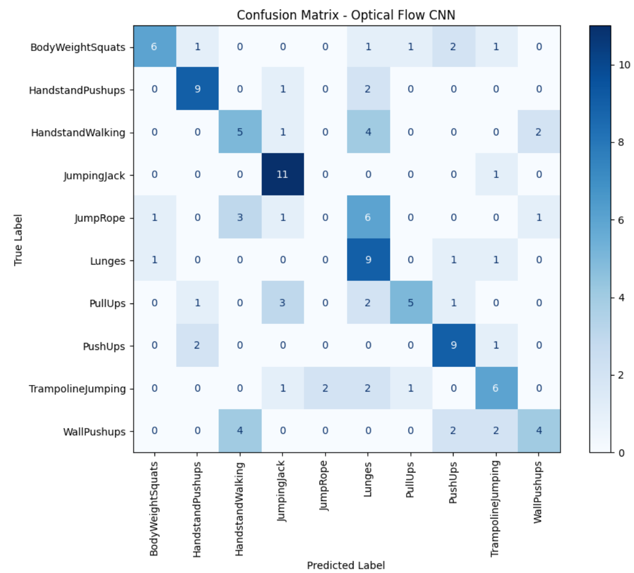

Figure 3. Confusion matrix for Dual Stream Network

As the results are not very satisfying, we tried also including pretrained models for both parts of the Dual Stream, specifically we used the ResNet18 [2] with the pretrained weights of ImageNet [1]. Using this, the results improved but not in a very significant way, obtaining a test accuracy of **62,5%** and a test loss of **1.5603**.

5. Conclusion

To sum up, we completed the project about physiscal exercises video classification successfully although the results in the last part where lower than expected. In general, the results of the first part where satisfaying achieving good metrics in most of the models trained. With this project we could fully understand and apply what we learnt about video classification from the different models we can use to more advanced models like the Dual Stream Network.

References

[1] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. 3

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *arXiv preprint arXiv:1512.03385*, 2015. 3

[3] Karen Simonyan and Andrew Zisserman. Two-stream convolutional networks for action recognition in videos. *arXiv*, 2014. 3